

Speaker notes (English)

Generated from presentation.html (17 slides incl. gag and appendix). Per slide: kicker, title, on-slide punch line, notes. Appendix slides and the two gag slides are uncouned.

Gag · Netscape: Yahoo!

(no notes)

Slide 1 · Explainability: / why did the model say that?

- Two days, **three models**: pneumonia classifier, lesion segmenter, report-writing VLM
 - Each gives **an answer**, none gives **a reason**
 - Talk: how we tried to get the reason out
 - ... and how often it turned out to be an **illusion**
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Slide 2 · Three questions, three models, one rule

Overview

1904: a horse called Clever Hans could do arithmetic. He was reading the questioner's face. / Every model we trained is a potential Clever Hans. The job is finding the face.

- Clever Hans 1904: no arithmetic, reads the face → every model a potential Hans
- Explainability ≠ one method; the question depends on the output
- Label → **“where?”**
- Mask → **“where is it wrong?”**
- Text → **“why?”** = hardest: the model happily gives a why unrelated to its decision
- Rule today: every explanation + **control** (change or remove something → does the explanation **survive?**)

Slide 3 · Where did it look?

Day 1 · classifier

The heat map is not wrong. It is just not saying much.

- Four images left to right: pneumonia right (TP), normal right (TN), normal called sick (FP), pneumonia missed (FN)
- **Grad-CAM**: last convolutional block with a grid, upsampled
- **Saliency**: each pixel → how much it moves the output; sharp, noisy
- **Integrated Gradients**: saliency averaged along a path from a black image
- Agree? **Only loosely**
- Grad-CAM **stays inside the thorax**: TP right lung + lower left field, FN upper left field; **frame edges never light up**
- Lungs or chest as a whole? → next slide
- Saliency / IG at 28 px: **salt and pepper**, roughly mediastinum and hila, no region to show a radiologist
- Take-away: heat-map precision **limited by the layer**, not by the disease

Slide 4 · Does it look at the lungs?

Day 1 · classifier

A good knife thrower misses on purpose. Our classifier misses the lungs by chance.

- A heat map alone has no “right place” → needs a definition
- Definition: **U-Net segments the lungs**; measure = share of heat inside the mask
- **1/3 of the attention on the lungs**, 1/3 of the image is lung → **chance level**
- Random network worse → training moved the gaze, but not into the lung fields
- TN and FP: heat on **mediastinum and heart**
- Hottest pixel inside the lungs: **23% of 624 images**
- At 28 px it learned **shape and brightness of the chest**, not the infiltrate
- Honesty: lung mask is itself a model, **Dice 0.76** vs. an independent segmenter → an average, not a per-image verdict

Slide 5 · The placebo test for heat maps

Day 1 · classifier

- **Adebayo et al. 2018**: destroy everything learned → does the explanation change? For popular methods barely
- Our test: trained net vs. random weights, correlation of the maps
- Correlation **near 1** = map follows the image edges, not the learning
- **Near 0** = trained model really looks somewhere specific
- Result: mean correlation **-0.03 → control passed**
- Random net paints the frame edges (left, bottom) for normal and FN; flat map for the TP
- Trained model: **never on the border**
- Run this control before showing any heat map to a clinician

Slide 6 · Where does it fail?

Day 2 · U-Net · control: look at the worst cases

Mean Dice 0.897. The mean is where the 22 failures hide.

- Segmentation: **no heat map needed**, the prediction is already a picture → honest explanation = **error map**
- Control: not the mean, the **worst cases**
- Mean Dice **0.897**, median 0.932; **22 of 392 below 0.7**
- Three worst = three stories:
- Left: **hair** → dark tangle painted as lesion
- Middle: annotation outlines the whole glass plate → **reference wrong**, not the model
- Right: **pale lesion**, almost no contrast → only the dark core found
- Not wrong at random, wrong **on a type of image**
- (If time) head/neck Dice **0.83 vs. 0.92** trunk – that is where the hair is

Slide 7 · Does it know when it is wrong?

Day 2 · U-Net · control: flip the image

Where the four answers differ, the errors sit. The hair blob glows. The wrong annotation does not.

- Same question four times: image **flipped four ways**, no retraining, four forward passes
- Map: **bright = the four answers disagree**
- Hair case left: false-positive blob **glows** → painted, but unsure
- Glass-plate case: **dark** → model confident; the annotation is wrong → disagreement **cannot** catch that
- 392 test images: disagreement vs. Dice correlate at **-0.36**
- Model roughly knows when it is on **thin ice**
- Not calibrated, not a probability, but **free**

Slide 8 · Why does it say that?

Day 2 · vision-language model · control: change the prompt

Same pixels, five prompts: 5 of 8 images change their label. The prompt decides, not the image.

- A language model gives the reason for free
- Control: **keep the pixels, change the prompt**, five times
- Pattern 1: model **over-calls** – 3 of 4 benign lesions “malignant” under almost every prompt
- Pattern 2: **“combined”** (few-shot + CoT + JSON) does the opposite: **3 of 4 melanomas benign**
- Why? Few-shot Example A = harmless mole → “melanocytic nevus” in **6 of 8** reports
- One melanoma: Example A reproduced almost **field by field** (symmetric, regular border, reassurance, confidence high)
- No sentences copied – **the answer copied**
- Naive prompt: asked for a report, wrote a **blank form**, field “patient name” in 6 of 8
- Which of the five is the explanation? **None** – five stories about the same pixels

Slide 9 · Same pixels, different patient

Day 2 · vision-language model · control: change the patient

The verdict never moved. The reasoning rewrote itself around it.

- Cannot open a language model → **change one input**, watch
- Two melanomas, same pixels; patient **25 / 75 / not stated**
- CoT + structured × 2 melanomas × 4 variants = **16 answers**
- Verdict **never moved: 16 of 16 malignant** → a control that passed, reported as such
- But the reasoning moved: at 25 **Spitz nevus**, at 75 **age as risk factor**
- Justification is **written around the answer**, not the other way round

Slide 10 · The colour of nothing

Day 2 · vision-language model · control: remove the image

Nothing to see: 10 of 14 times "benign, reassure". Red: surgery.

- Last control: **take the image away** – grey square + the melanoma's clinical context
- Answer: quality poor, no structures, no colours ... impression **benign**, recommendation **reassure**
- Answer to nothing, or only to grey? → **seven colours**
- Grey, black, white, skin, dark brown: benign, reassure, every time
- **Red: suspicious, excision** – DD amelanotic melanoma, angioma, BCC; despite “no structures, low confidence”
- **Light brown**: suspicious for the 80-year-old woman on the head, benign for the 65-year-old man on the trunk → **metadata flip the verdict**
- Real lesions: age never moved it; empty image: the prompt is all the model has
- Seeing nothing and saying “harmless” = **most dangerous default** in dermatology
- 14 calls, about one minute

Slide 11 · What we take home

Day 1 + Day 2

- **Ghassemi, Oakden-Rayner, Beam**: explainable AI in health care = “false hope”
- Not because the methods are useless, but because they are shown **without controls**
- Every control we ran: **minutes**
- Thank you

Appendix · Appendix

(no notes)

Appendix · Does it separate the classes internally?

Appendix · Day 1 · classifier

- Global view: **two clouds**, normal left, pneumonia right
 - In between a **mixed band** = false positives and negatives
 - The model has an internal notion of the two classes
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Appendix · Same pixels, eight models

Appendix · Day 2 · VLM · control: change the model

Eight second opinions. Not one of them is a second opinion on the image; each is a first opinion on its own training data.

- A Qwen problem? Same 8 images, same structured prompt, **7 other models** on Bedrock
- Nobody above **5 of 8**; the mistakes are **not the same**
- Nova, Pixtral, Grok: **every benign lesion malignant**
- Sonnet 5: the opposite, **3 of 4 melanomas benign**; once “impression benign” + “recommendation excision” in the same JSON
- Errors in different directions → **majority vote no better** than the best single model
- For explainability: the label is **not a property of the pixels** – all 8 images change label when only the model changes
- DD for the 15-year-old's nevus: **5 models, 5 diseases** (dermatofibroma ... melanoma in situ), same calm voice
- Model choice = **hidden hyperparameter** of the explanation

Appendix · Who agrees with whom?

Appendix · Day 2 · VLM · control: ask twice

The models that agree most are the ones that say "malignant" to everything. Agreement is not accuracy.

- Same data as a chord diagram: wide band = same label on many of the 8 images
- Widest bands: **Nova Pro, Pixtral, Qwen** – 7 or 8 of 8
- Reason: they say malignant to almost everything → always saying the same means always agreeing = **shared bias**, not insight
- Sonnet 5 at the other end: agrees on 2 images, errs in the opposite direction
- A committee does not average out the errors, it averages **the biases**
- Box at the bottom: everything run **twice** with identical settings → Sonnet 5 and Nova each lost a hit, majority **6 → 5**
- An explanation that changes when you ask again was never an explanation of the image: **a sample**

Appendix ·

(no notes)